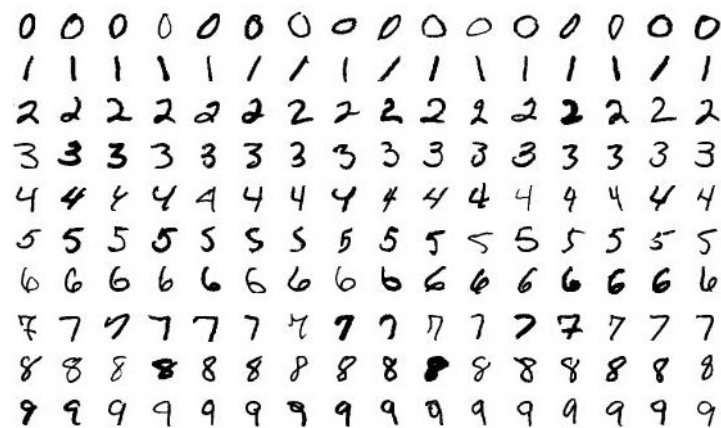


Exercise 1a - 19th of February 2024

First Classification Task with KNN

Deadline: 4th of March 2024 (end of day)

The aim of this exercise is to implement your first classifier and apply it to the popular MNIST dataset,¹ which contains images of handwritten digits.



In this exercise, your task is to implement the KNN algorithm and test it on the dataset.

Data and Features

On Ilias, you'll find the zip file with two csv files (train.csv, test.csv). The csv files contain the labels (drawn number) and pixel values for the 28×28 pixels in the greyscale images, with values in the range of $[0, 255]$. The first column in the csv file represents the class label and the remaining 784 values the pixel values. The images are represented as 1D arrays, so you need to reshape the data if you want to display the images on the screen.

The pixel values should be used directly as the features. You don't need to compute any other features (although you certainly can if you like to).

¹<http://yann.lecun.com/exdb/mnist/>

KNN

Create your own implementation of the KNN algorithm (see lecture notes). Classify the entries of the dataset using your KNN implementation. Experiment with different values of k . Aim for a fast implementation as the dataset is of considerable size.

Implement at least two different distance metrics (e.g. Euclidean and Manhattan) and evaluate which one works best for you.

Expected Submission

Hand in your solutions on Ilias `exercises/1a-KNN`: `hand-in` with:

- Source Code of your implementation.
- Short report with your results:
 - Accuracy of your classification for $k = \{1, 3, 5, 10, 15\}$ with the chosen distance metrics.
 - *Optional*: Confusion matrix